



Bayesian Scoring Using PROCs MCMC and IML

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The MCMC Procedure

- *PROC MCMC* is a general purpose simulation procedure that uses Markov chain Monte Carlo (MCMC) techniques to fit a wide range of Bayesian models.
- It requires the specification of a likelihood function for the data and a prior distribution for the parameters.
- It enables you to analyze data that have any likelihood or prior distribution as long as they are programmable using SAS DATA step functions.

Necessary Information for PROC MCMC

Specify a joint probability distribution for observed data

D and model parameter θ in two steps:

1. The conditional density of data D given the parameter θ (which is $P(D|\theta)$);
and
2. The marginal density of the parameter θ (which is $P(\theta)$):

Above, two steps generate the conditional density of θ given the data D:

$$P(\theta | D) = \frac{P(D | \theta)P(\theta)}{P(D)}$$

using Bayes' rule.

PROC MCMC Statements

- You declare the parameters in the model and assign the starting values for the Markov chain with PARMS statements.
- You specify prior distributions for the parameters with PRIOR statements.
- You specify the likelihood function for the data with the MODEL statement.

Simple Linear Regression

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad \varepsilon_i \sim N(0, \sigma^2)$$

for subjects $i=1,2,\dots,n$,

The above model can equivalently be expressed as

$$Y_i \sim \text{normal}(\beta_0 + \beta_1 X_i, \sigma^2)$$

for subjects $i=1,2,\dots,n$

Simple Linear Regression

The prior distributions for the three parameters are

$$\pi(\beta_0) = \textit{normal}(0, \text{var} = 1e6)$$

$$\pi(\beta_1) = \textit{normal}(0, \text{var} = 1e6)$$

$$\pi(\sigma^2) = \textit{inversegamma}(\textit{shape} = 2.001, \textit{scale} = 1.001)$$

Simple Linear Regression

```
proc mcmc data=slr seed=27513;  
  parms beta0 0 beta1 0;  
  parms sigma2 1;  
  prior beta0 beta1 ~ normal(mean=0,  
    var=1e6);  
  prior sigma2 ~ igamma(shape=2.001,  
    scale=1.001);  
  mu=beta0 + beta1*X1;  
  model Y ~ normal(mu, var=sigma2);  
run;
```

PROC MCMC Syntax

PROC MCMC *options*;

PARMS *parameters and starting values*;

BEGINCNST;

Programming Statements;

ENDCNST;

BEGINNODATA;

Programming Statements;

ENDNODATA;

PRIOR *parameter ~ distribution*;

MODEL *variable ~ distribution*;

RANDOM *random effects specification*;

PREDIST <'label'> OUTPRED=SAS-data-set <options>;

RUN;

PREDDIST Statement

- The PREDDIST statement creates a new SAS data set that contains random samples from the posterior predictive distribution of the response variable.
- The posterior predictive distribution can often be used to check whether the model is consistent with the data.
- The PREDDIST statement works only on response variables that have standard distributions, and it does not support either the GENERAL or DGENERAL functions.

Frequentist versus Bayesian Approach to Scoring

SCORE Statement in Frequentist Procedures

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 \dots$$

PREDDIST Statement in PROC MCMC

$$\hat{Y}_i = \hat{\beta}_{0_i} + \hat{\beta}_{1_i} X_1 + \hat{\beta}_{2_i} X_2 \dots$$

where i = 1 to number of saved iterations



Code Demonstration

A series of horizontal bars of varying lengths and colors (teal, blue, and dark blue) are arranged on the left side of the slide, creating a decorative, layered effect.

Thanks for attending.

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